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PhD in Machine Learning, dissertation submitted September 2026, defence expected December 2026

My research focuses on **steered generative modeling** and **graph-structured learning**, with applications to **neural architecture search** and **anomaly detection** on **dynamic networks**:

- RL-steered **graph diffusion** for **neural architecture search** on directed acyclic graphs (DGPO): matches the NAS-Bench-201 optima on all three tasks (by benchmark lookup); pretrained on 7% of NAS-Bench-101 (architectures below 87% accuracy only), the best generated architecture among evaluation draws lands 7.27 pp above that ceiling. Bidirectional-control ablation: inverse reward drives accuracy to 9.5% (chance), validating directional control of the generator. *IJCNN 2026.*
- Unsupervised **anomaly detection on temporal graphs**: TGN-SVDD reaches ROC-AUC 0.994 - 0.999 vs 0.268 - 0.690 for vanilla TGN on CIC-IDS2017 (ICANN 2023); its probabilistic extension RTGN-SVDD raises ROC-AUC 0.574 → 0.844 and 0.561 → 0.880 vs the deterministic model with random noise events injected at 50% of normal traffic (ESANN 2024).
- Multi-objective **AutoML** pipelines for Pareto-optimal **accuracy-sparsity trade-offs** for interpretable sparse models, evaluated on open-source cybersecurity datasets (TPOT, Optuna, NSGA-II).

Experience

Nov 2020 – Present **Doctoral Researcher, Machine Learning**, *Bielefeld University*, Bielefeld, Germany
- Research on generative AI for neural architecture search (graph diffusion + RL), anomaly & OOD detection with AutoML, GNNs, and probabilistic methods.
- Published 6 papers at international conferences and journals (5 as first author).
- Ran multi-seed training and evaluation campaigns on SLURM GPU clusters (single-GPU, ∼115 GPU-hours for DGPO).
- Built AI-augmented research workflows (Claude Code, Codex) to automate experimental pipelines, from implementation through systematic evaluation.

Oct 2023 – Sep 2025 **Tutor & Supervisor**, *Bielefeld University*, Bielefeld, Germany
Applied Optimization (Oct 2023 – Mar 2024; Oct 2024 – Mar 2025): designed exercises, supervised and coordinated fellow tutors. AI Act in ML Seminar (Apr 2024 – Sep 2024; Apr 2025 – Sep 2025): led seminar on AI regulation; supervised student projects.

Sep 2011 – Jul 2018 **Founder & CEO, Baltik Bread**, *Bakery Manufacturing & Mobile Retail*, Novosibirsk, Russia
- Co-founded and scaled a dual manufacturing + retail operation from zero; 500k - 600k EUR investment; full ROI within 4 years.
- Grew to ∼1M EUR annual revenue, ∼2,500 units/day across ∼40 SKUs with 90%+ sell-through; built and managed a 25-person team.
- Data-driven location optimization using empirical sales analysis; built competitive advantage through systematic testing.

Education

Nov 2020 – Sep 2026 **PhD in Machine Learning**, *Bielefeld University*, Bielefeld, Germany
Dissertation submitted Sep 2026, defence expected Dec 2026. Dissertation: *Graph-Based Machine Learning: From Modeling Structured Data for Security to Generating Neural Architectures*. Supervisor: Prof. Barbara Hammer, Machine Learning Group (HammerLab).

Oct 2018 – Mar 2021 **M.Sc. Data Science**, *Bielefeld University*, Bielefeld, Germany
Thesis: Automated Machine Learning (AutoML) technologies for feature selection.

Sep 2007 – Jun 2010 **Master of Economics in Mathematical Methods of Economic Analysis**, *Novosibirsk State University*, Novosibirsk, Russia

Sep 2003 – Jun 2007 **B.Sc. Physics**, *Novosibirsk State University*, Novosibirsk, Russia

Publications

- **A. Liuliakov**, L. Hermes, and B. Hammer. "DGPO: RL-Steered Graph Diffusion for Neural Architecture Generation." In: *IJCNN 2026 (WCCI)*. [arXiv] [code]
- **A. Liuliakov**, A. Schulz, L. Hermes, and B. Hammer. "Noise Robust One-Class Intrusion Detection on Dynamic Graphs." In: *ESANN 2024*. [arXiv] [code]
- **A. Liuliakov**, A. Schulz, L. Hermes, and B. Hammer. "One-Class Intrusion Detection with Dynamic Graphs." In: *ICANN 2023*. [arXiv] [code]
- L. Hermes, **A. Liuliakov**, and M. Schilling. "Graph Learning by Dynamic Sampling." In: *IJCNN 2023*.
- **A. Liuliakov**, L. Hermes, and B. Hammer. "AutoML Technologies for the Identification of Sparse Classification and Outlier Detection Models." In: *Applied Soft Computing*, 2023.
- **A. Liuliakov** and B. Hammer. "AutoML Technologies for the Identification of Sparse Models." In: *IDEAL 2021*.

Service

Session chair	'AI for Finance and Economics', IJCNN 2026 (IEEE WCCI, Maastricht)
Peer reviewing	ESANN 2024 (1), IJCNN 2024 (1), ESANN 2026 (2), IJCNN 2026 (2) papers
Teaching	Sole seminar instructor, 'Implications of the AI Act in Machine Learning', Bielefeld University, SoSe 2024 and SoSe 2025

Skills

ML & AI	Generative AI, Diffusion Models, Reinforcement Learning, Graph Neural Networks, Deep Learning, AutoML, Neural Architecture Search, Anomaly Detection, OOD Robustness, Probabilistic Modeling
Programming	Python, PyTorch, PyTorch Geometric, PyTorch Lightning, Optuna, scikit-learn, NumPy, Pandas, R, JavaScript, HTML, CSS
Platforms	Linux clusters, Slurm, Docker, W&B, Git, GitHub, VS Code
Engineering	Public research repos: DGPO (MIT, Makefile one-command reproduction, pinned requirements); rtgn_svdd (pyproject, pytest tests)
AI Tools	Claude Code, OpenAI Codex, GitHub Copilot: agentic software development with spec-first plans, test-driven increments and cross-model code review; the DGPO codebase and experiment pipeline (training, RL fine-tuning, NAS evaluation, SLURM cluster runs) were largely built this way
Research Methods	Mathematical Modeling, Statistics, Applied Optimization, Experimental Design, Scientific Writing

Languages

English	Full professional proficiency (C1)
German	Limited working proficiency (B1)
Russian	Native
French	Elementary (A1)